Hustle for Cash:
Evidence from Nigeria's Naira Crunch

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Abstract

When Nigeria's central bank abruptly withdrew currency in late 2022 and capped cash withdrawals, the resulting shortage, the *naira crunch*, created a sharp, unanticipated cash supply shock. This paper exploits cross-state variation in pre-shock digital payment adoption, measured eleven days before the policy announcement, to estimate the real costs of the crunch and the insulating role of financial infrastructure. Using enterprise modules from the Nigeria Living Standards Phone Survey (1,842 household enterprises across three rounds), with the parallel-trends assumption validated through a 45-month pre-shock nighttime-light panel covering all 774 Nigerian local government areas, enterprises in states one standard deviation above the pre-shock fintech mean were 1.7 percentage points more likely to remain in operation at peak crunch and 7.7 percentage points more likely to survive fourteen months later. Employment effects are also large: 0.29 additional enterprise workers per standard deviation of fintech at peak crunch. Effects are concentrated in rural and small enterprises. The results identify digital financial infrastructure as a macroeconomic stabiliser against cash supply disruptions, with implications for central bank currency operations in low-income economies.

JEL codes: E41, E52, G23, O16, O55

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1 Introduction

In late October 2022, Nigeria's central bank announced the withdrawal and redesign of its currency notes, together with strict daily caps on cash withdrawals. By January 2023, a rolling shortage of the replacement notes had produced hour-long queues outside commercial banks, paralysed point-of-sale terminals, and darkened markets where almost every transaction had always settled in cash. Prices for essential goods spiked and some small enterprises closed their doors. In February 2023 the Supreme Court ordered the old notes to remain legal tender, and the central bank progressively released more new currency over the following months. The episode, the *naira crunch*, lasted approximately six months before cash supply normalised.

The crunch was an abrupt, policy-induced cash supply shock where the stock of physical currency available for transactions fell sharply while the demand for cash-settled transactions did not. Its severity, however, varied substantially across Nigeria's thirty-six states and the Federal Capital Territory (FCT), for a reason not immediately obvious at the time as some states had higher pre-shock adoption of digital payment technology. In those states, enterprises and households could substitute to mobile money wallets, USSD bank transfers, and fintech applications when ATMs ran dry. In states where most transactions still settled exclusively in cash, limited substitute existed.

This paper exploits that cross-sectional heterogeneity to estimate the real costs of a cash supply shock and the extent to which digital financial infrastructure acts as a buffer. The identification relies on three features of the episode. First, the CBN announcement on October 26, 2022 was genuinely unanticipated, making the pre-shock distribution of fintech adoption a predetermined variable uncontaminated by anticipation of the shock. Second, that distribution is measured using the Nigeria Living Standards Phone Survey (NLPS) Round 6 which was conducted on October 15, 2022, eleven days before the announcement, providing a state-level index of digital payment penetration constructed from four digital-payment service categories.[1] Third, the parallel-trends assumption is validated using monthly composites of nighttime-light radiance from the NASA VIIRS Black Marble satellite product for all 774 Nigerian local government areas (January 2019 to September 2022), providing 45 consecutive pre-shock months over which high- and low-fintech states can be shown to have followed statistically identical trajectories prior to the crunch. Section 6 details the identification

[1]The index uses four digital-payment categories: mobile money, commercial bank accounts, mobile and USSD banking, and mobile banking applications. Four further categories (pension/insurance, securities, microfinance, and credit cards) are excluded because their penetration is below 4% in all states and they do not constitute a meaningful cash substitute during the crunch; see Appendix Table 15 for robustness to alternative index definitions. Section 5 details the construction.

strategy and pre-trend validation.

The primary evidence comes from firm-level data.and the main findings are as follows. Enterprises in states one standard deviation above the pre-shock fintech mean were 1.7 percentage points more likely to remain in operation at the February 2023 peak crunch and 7.7 percentage points more likely to remain in operation fourteen months later in April 2024. Employment was also significantly insulated as high-fintech states had 0.29 more enterprise workers during the peak crunch. Effects are concentrated in rural firms and small enterprises below-median size, the units most dependent on daily cash flow.

These results contribute to three literatures. First, they extend Chodorow-Reich et al. (2019) to the mobile-money studies. Chodorow-Reich et al. (2019) shows that Indian districts with fewer bank branches lost more economic activity during the 2016 demonetisation; the relevant margin in Sub-Saharan Africa is not bank branches, but digital payment technology. This paper identifies a *digital payment substitution channel* that the India paper's design, which relies on bank-deposit variation, cannot isolate. The magnitude of the enterprise-survival estimates implies that closing the fintech adoption gap between the 10th- and 90th-percentile Nigerian states would have reduced firm exit at the peak crunch by approximately 5 percentage points, a substantial share of the 16-percentage-point decline in enterprise activity observed between Round 5 and Round 7.

Second, this paper provides evidence for a digital-payment substitution mechanism emphasized in the mobile-money literature. By enabling transactions and transfers to be executed without physical currency, digital payment technologies reduce reliance on cash and may attenuate the effects of cash-supply disruptions (Jack and Suri, 2014). The present study quantifies the extent of this insulation in LMICs using plausibly exogenous pre-determined variation in fintech adoption.

Third, the analysis contributes to the emerging methodological literature on identifying monetary transmission in data-sparse economies by leveraging alternative proxies for economic activity (see Henderson et al., 2012; Donaldson and Storeygard, 2016). High-frequency satellite imagery combined with granular phone survey data can recover real monetary transmission at subnational resolution where the formal national accounts provide limited quarterly or monthly benchmarks. The satellite–survey combination is directly replicable for other African central bank episodes.

Section 2 describes the naira crunch and Nigeria's payment landscape. Section 3 surveys the related literature. Section 4 presents the theoretical framework. Section 5 introduces the data. Section 6 presents the research design and validates the identifying assumption using the 45-month NTL pre-trend window. Section 7 reports the firm-level results. Section 8 examines longer-run effects using the 2025 World Bank Enterprise Survey and the April 2024

NLPS round. Section 9 concludes with policy implications.

2 The Naira Crunch: Institutional Context

2.1 The Policy Shock

On October 26, 2022, the Central Bank of Nigeria (CBN) announced the redesign of its three highest-denomination banknotes (₦200, ₦500, and ₦1,000) and set a January 31, 2023 deadline for old notes to be deposited and replaced (Central Bank of Nigeria, 2022). Daily ATM withdrawals were capped at ₦20,000 (approximately $43 at the prevailing rate) and over-the-counter withdrawals at ₦100,000 per week. The stated rationale was to combat counterfeiting, reduce the volume of currency held outside the banking system, and accelerate the adoption of digital payments in support of the CBN's broader cashless-economy agenda.

The roll-out of new notes proved severely inadequate to demand. By mid-January 2023, queues outside commercial banks and ATMs stretched for hours across major Nigerian cities. Reports from the field and media documentation consistently describe near-total unavailability of new notes in rural areas, with POS machines dysfunctional and markets at a standstill.[2] The naira crunch was by this point a genuine constraint on economic activity, not merely an inconvenience.

In February 2023 Supreme Court of Nigeria ruled that old notes must remain legal tender until December 2023, providing partial relief. Concurrently, the CBN accelerated distribution of new currency. By mid-2023 most commercial channels had normalised, though NLPS survey evidence (Round 11, April 2024) and the 2025 WBES both indicate persistent behavioural shifts in payment habits.

Table 1 summarises the shock timeline.

[2]The *NBS Enterprise Telephone Survey*, *SBM Intelligence* rapid surveys, and contemporaneous New Agency of Nigeria *NAN* wire reports provide granular documentation. These sources are not used for identification; they motivate the timing of the shock indicators.

Table 1. Naira Crunch: Policy Timeline

Date	Event
October 26, 2022	CBN announces naira redesign; caps daily ATM withdrawals at ₦20,000; sets January 31 deadline for old notes.
November–December 2022	Transition period: old notes recalled, new notes printed but distribution severely below demand. Early shortages in smaller cities and rural areas.
January–February 2023	*Peak crunch*: near-complete ATM dry-outs in many states; hour-long bank queues; POS network failures; informal market disruption.
February 8, 2023	Supreme Court orders old notes remain legal tender; CBN given three weeks to increase note supply.
March–June 2023	Gradual recovery as new note circulation increases; cash availability improves progressively by state.
July 2023 onward	Post-crunch: cash supply normalised in most states; some permanent shift toward digital payment habits documented in subsequent surveys.

Notes: Timeline constructed from CBN official circulars, Supreme Court records, and National Bureau of Statistics rapid-survey documentation. Shock indicator periods follow this timeline exactly; see Section 6 for period definitions.

2.2 Nigeria's Payment Landscape

Understanding why fintech adoption creates differential exposure requires appreciating Nigeria's payment infrastructure at the time of the shock. According to the 2021 *Global Findex*, approximately 45 percent of Nigerian adults held a formal bank account, well below South Africa (85%). Bank branch density is heavily concentrated in Lagos, Abuja, Port Harcourt, and Kano; many northern and rural states have fewer than two commercial bank branches per 100,000 inhabitants (International Monetary Fund, 2022).

Against this low-baseline context, fintech adoption expanded rapidly between 2018 and 2022. Three developments are particularly relevant. First, mobile money operators (notably OPay and PalmPay) scaled their agent networks aggressively, reaching peri-urban and rural areas bypassed by traditional banking. Second, the CBN's mandatory tier-1 Know-Your-Customer (KYC) relaxation in 2018 lowered barriers to opening mobile wallets. Third, USSD banking (dialling a short code to transact without internet) penetrated widely through the mobile network operator infrastructure.

A concern is whether these CBN-directed policies render fintech adoption endogenous to the 2022 crunch that if states that responded most to the 2018 KYC relaxation were also anticipating a cashless policy, the pre-shock fintech index might proxy for policy expectations rather than exogenous infrastructure variation. Three features of the episode address this concern. First, the specific form of the 2022 shock, a forced currency redesign combined with withdrawal caps, was not a predictable consequence of the 2018 KYC relaxation; the two episodes are legally and operationally distinct, separated by four years. Second, the NLPS Round 6 financial service module, conducted 11 days before the announcement, directly records *realised* adoption, not plans or expectations. Also, the comparison of Round 6 to Round 7 (February 2023) shows significant changes in financial service usage, confirming agents had not positioned in advance. Third, cross-state variation in fintech take-up shows differential adoption of mobile infrastructure, driven by geography, agent-network rollout costs, and pre-existing mobile penetration, rather than differential anticipation of currency policy, as the geographic analysis in Section 6.4 shows.

By October 2022, the NLPS Round 6 financial services module (conducted 11 days before the CBN announcement) recorded the following state-level averages. 14.7 percent of households held a commercial bank account; 86.2 percent had used mobile phone or USSD banking; 5.9 percent held a mobile money account; and 5.1 percent used a mobile banking application. The composite fintech index, the simple average of the four digital-payment categories, ranged from 19.8 % (Oyo) to 40.5 % (Taraba) across states, providing the cross-sectional variation the identification exploits.

States with low fintech penetration are not simply poorer. Appendix Table 6 shows that the state-level fintech index is uncorrelated with pre-2022 nighttime light intensity ($r = -0.19$) and with pre-trend NTL growth ($r = 0.10$), supporting the parallel-trends assumption formalised in Section 6.

3 Related Literature

This paper contributes to four bodies of work as it relates to the economic effects of cash supply shocks, the role of digital payment technology in developing economies, the theory of money and payment systems, and the use of satellite nighttime lights as a high-frequency economic indicator.

3.1 Cash Supply Shocks and Currency Policy

The empirical literature on cash supply shocks is anchored by India's November 2016 demonetisation, when the government abruptly invalidated 86 percent of the currency in circulation.

Chodorow-Reich et al. (2019) exploit geographic distribution of demonetized versus newly printed bank notes across India's districts, and find that districts with fewer bank branches and higher cash reliance experienced significantly larger declines in economic activity, measured by nighttime lights and formal employment. In sub-Saharan Africa specifically in Nigeria, a relevant margin of financial access is digital payment technology that operates independently of physical branch infrastructure. This motivates the focus of this study on a fintech-based treatment variable and isolates a *digital payment substitution channel*.

Lahiri (2020) provides a complementary macroeconomic accounting of the 2016 episode, documenting that the GDP cost was concentrated in the informal sector and in the months immediately following the shock. Rogoff (2015) argues that the aggregate macroeconomic benefits of reducing paper currency use are substantial. Evidence from this current study points to an important asymmetry where reducing cash dependence through digital payment adoption is welfare-improving, but an abrupt withdrawal of currency without sufficient digital infrastructure to substitute imposes large real costs concentrated in the least financially included segments of the economy.

3.2 Digital Payments and Financial Inclusion

The literature has established the welfare effects of mobile money and digital payment technology in low-income settings. Jack and Suri (2014) provide the foundational causal evidence from Kenya's M-Pesa system, showing that mobile money enables households to share risk more effectively and smooth consumption against income shocks. Suri and Jack (2016) extend this to long-run poverty outcomes, finding that mobile money adoption increased per-capita consumption and reduced poverty, with larger effects for female-headed households. Mbiti and Weil (2016) finds that increased use of M-Pesa lowers the propensity to use informal savings mechanisms and raises the likelihood of being banked.

In West Africa, Aker et al. (2016) evaluate a mobile-money cash transfer programme in Niger and find significant reductions in transaction costs and improvements in household welfare relative to traditional payment channels. These studies establish that digital payment access provides direct welfare benefits through risk-sharing and consumption smoothing. The contribution of this study is to identify digital payment infrastructure as a stabiliser that insulates the real economy from disruptions in the supply of physical currency. This channel operates at the state level rather than the household level and is distinct from the individual-level welfare gains that had been shown in earlier studies.

Financial inclusion literature, Beck et al. (2007), Allen et al. (2016), Burgess and Pande (2005), had shown that access to formal financial services promotes household welfare and firm productivity through credit access, risk management, and reduced transaction costs. The naira crunch episode allows us to estimate the value of financial infrastructure through its performance under stress, when infrastructure serves as a substitute for a disrupted payments medium.

3.3 Monetary Theory and Payment Technology

The theoretical positioning for a digital-payment substitution channel can be traced to the cash-in-advance (CIA) tradition originating with Lucas and Stokey (1987). In these models, a subset of transactions must be financed with liquid balances held prior to purchase. Consequently, an unexpected reduction in the availability of currency tightens liquidity constraints, restricts transactions, and reduces economic activity. Building on Alvarez and Lippi (2009) that show that financial innovation lowers the demand for transaction balances by reducing reliance on cash, access to digital payment technologies can be interpreted as relaxing transaction-related liquidity constraints. Enabling some transactions to be settled electronically rather than with physical currency, digital payment systems reduce dependence on cash as the sole medium of exchange. In the context of a cash shortage, agents with greater access to digital payment technologies should therefore be better able to maintain transactions and economic activity.

The empirical design of this study operationalises this mechanism at the subnational level. States with higher pre-shock fintech adoption have, in aggregate, a lower dependence on physical currency for transaction settlement; when cash supply contracts, the real CIA constraint binds less severely in these states. Section 4 formalises this intuition and derives the testable predictions that the empirical sections evaluate.

3.4 Satellite Nighttime Lights as Economic Measurement

High-frequency satellite data have become a complement to official statistics where administrative data are unavailable at monthly or subnational resolution. Henderson et al. (2012) establish the foundational result that nighttime-light radiance reliably proxies economic activity, especially for countries with weak national accounts. Donaldson and Storeygard (2016) survey applications in economics ranging from road infrastructure to conflict detection. Gibson et al. (2020) discuss appropriate uses and limitations, including sensitivity to cloud cover, overglow, and saturation in dense urban areas. Brüderle and Hodler (2018) validate nighttime lights as a proxy for local human development across Africa, supporting their use in the sub-Saharan context.

For this paper, a contribution of the NASA VIIRS Black Marble product is not as a primary outcome but as an identification tool. The 45 pre-shock monthly observations of LGA-level luminosity (January 2019 to September 2022) permit a test of parallel trends in the state-level fintech treatment variable, validating the identifying assumption for the firm-level specifications in Section 7.

4 Theoretical Framework

This section develops a formal model of enterprise behaviour under a cash supply shock to derive testable predictions. The framework embeds heterogeneous payment technologies into a cash-in-advance (CIA) model of Lucas and Stokey (1987).

4.1 Environment

Enterprises. A continuum of enterprises is indexed by $i \in [0,1]$, each located in state $s \in \{1,\dots,S\}$. Enterprise i produces a single good with the production function

$$y_i = A_i\,\ell_i^{\alpha}, \qquad \alpha \in (0,1), \tag{1}$$

where $A_i > 0$ is enterprise-specific productivity and $\ell_i \geq 0$ is labour input hired at competitive wage $w > 0$. The enterprise sells output at price p (normalised to one) and maximises profit net of a fixed operating cost $\kappa_i > 0$:

$$\pi_i \;=\; A_i\ell_i^{\alpha} - w\ell_i - \kappa_i. \tag{2}$$

Payment technology. Wages must be paid before output is sold, a timing friction generating a transactions demand for a means of payment. Let $\phi_{is} \in [0,1)$ denote the share of the wage bill that enterprise i can settle through digital channels (mobile money, USSD transfers, fintech applications). The remainder $(1-\phi_{is})$ must be settled in physical cash. Parameterise $\phi_{is} = \phi(F_s,\nu_i)$, where $F_s \geq 0$ is the state-level fintech penetration index and ν_i is enterprise-specific idiosyncratic digital access, with $\partial\phi/\partial F_s > 0$.

Cash-in-advance constraint. The enterprise must hold sufficient physical currency before hiring:

$$m_{is} \;\geq\; (1-\phi_{is})\,w\,\ell_{is}, \tag{3}$$

where $m_{is} \leq \bar{m}_{is}$ is the stock of physical cash available to the enterprise, bounded above by local cash supply $\bar{m}_{is}$ that is proportional to the state aggregate M_{st}.

4.2 Unconstrained and Constrained Optima

When the CIA constraint (3) does not bind, the enterprise chooses the first-best labour input:

$$\ell_i^{\mathrm{FB}} \;=\; \left(\frac{\alpha A_i}{w}\right)^{1/(1-\alpha)}. \tag{4}$$

The first-best profit is $\pi_i^{\mathrm{FB}} = (1-\alpha)\,A_i\,(\ell_i^{\mathrm{FB}})^{\alpha} - \kappa_i$.

When the CIA constraint binds—because local cash supply $\bar{m}_{is}$ falls short of $(1-\phi_{is})\,w\,\ell_i^{\mathrm{FB}}$—the enterprise is rationed to:

$$\ell_i^{*} \;=\; \frac{\bar{m}_{is}}{(1-\phi_{is})\,w}, \tag{5}$$

and constrained profit is

$$\pi_i^{*} \;=\; A_i\left(\frac{\bar{m}_{is}}{(1-\phi_{is})\,w}\right)^{\alpha} - \bar{m}_{is}/(1-\phi_{is}) - \kappa_i. \tag{6}$$

The CIA constraint binds if and only if

$$\bar{m}_{is} \;<\; (1-\phi_{is})\,w\,\ell_i^{\mathrm{FB}}. \tag{7}$$

4.3 The Cash Supply Shock

Prior to the shock (October 2022), state cash supply is $M_{s,\mathrm{pre}} = \bar{M}$ for all s, with $\bar{M}$ high enough that the CIA constraint does not bind for any enterprise. The naira crunch reduces available cash to $M_{s,\mathrm{crunch}} = \bar{M} - \Delta_s$, where $\Delta_s > 0$ for all s. For sufficiently large Δ_s, the constraint (3) binds.

Exit decision. Enterprise i exits if constrained profit falls below zero:

$$\pi_i^{*} < 0 \quad\Longleftrightarrow\quad \bar{m}_{is} < \underline{m}_i(\phi_{is}), \tag{8}$$

where $\underline{m}_i(\phi_{is})$ is an exit-threshold cash level that is *decreasing* in ϕ_{is}, as shown in Proposition 1 below.

4.4 Proposition: Digital Payment Insulation

Proposition 1 (Digital Payment Insulation)**.** *Suppose the* CIA *constraint* (3) *binds at the crunch. Then:*

(i) ***Employment.*** *Constrained employment ℓ_i^* is strictly increasing in the digital payment share ϕ_{is}:*

$$\frac{\partial \ell_i^*}{\partial \phi_{is}} = \frac{\bar{m}_{is}}{(1-\phi_{is})^2\, w} > 0.$$

(ii) ***Survival.*** *The exit threshold $\underline{m}_i(\phi_{is})$ is strictly decreasing in ϕ_{is}: a higher digital payment share expands the set of cash-supply realisations under which the enterprise survives.*

(iii) ***Heterogeneity.*** *The employment gain from a marginal increase in ϕ_{is} is strictly increasing in the cash supply contraction Δ_s:*

$$\frac{\partial^2 \ell_i^*}{\partial \phi_{is}\, \partial \Delta_s} > 0.$$

Proof. Part (i) follows directly from differentiating (5) with respect to ϕ_{is}.

For part (ii), differentiate the exit condition in (8) implicitly. Constrained profit $\pi_i^*(\bar{m}_{is}, \phi_{is})$ satisfies $\partial \pi_i^*/\partial \bar{m}_{is} > 0$ (more cash raises output and profit) and $\partial \pi_i^*/\partial \phi_{is} > 0$ (higher digital share raises constrained employment and profit). Setting $\pi_i^* = 0$ and applying the implicit function theorem gives $\mathrm{d}\underline{m}_i/\mathrm{d}\phi_{is} = -(\partial \pi_i^*/\partial \phi_{is})/(\partial \pi_i^*/\partial \bar{m}_{is}) < 0$.

Part (iii) follows from the cross-derivative of (5): since $\bar{m}_{is} = \bar{M} - \Delta_s$ decreases in Δ_s, $\partial \ell_i^*/\partial \phi_{is} = \bar{m}_{is}/[(1-\phi_{is})^2 w]$ falls as Δ_s rises, but the *level* effect of ϕ_{is} on employment is larger in absolute terms when Δ_s is large, because the binding constraint is tighter. □

4.5 Testable Predictions

The three parts of Proposition 1 map directly to the empirical specifications.

Prediction 1. Part (i) predicts a positive coefficient on $\textit{Fintech}_s \times \mathbf{1}[\text{peak crunch}]$ in the enterprise employment regression (Section 7.1).

Prediction 2. Part (ii) predicts a positive coefficient in the enterprise survival linear probability model; and that the survival advantage is larger when cash supply falls further (tested across shock phases in Section 7.1).

Prediction 3. Part (iii) predicts that the insulation effect is larger for enterprises with higher baseline cash dependence—specifically small and rural enterprises for whom the CIA constraint is most likely to bind. Tested in Section 7.2.

Prediction 4. Aggregating across enterprises within a state, a higher average $\phi_s = \mathbb{E}_i[\phi_{is}|s]$ reduces state-level output loss. This is validated using the nighttime-light pre-trend test in Section 6.3.

4.6 Theory Discussion

The insulation channel operates through *payment medium substitution*, not credit access. Enterprise i is not assumed to borrow more or have better collateral in high-fintech states; it simply settles a larger fraction of its wage bill without physical cash, relaxing the binding constraint (3). This distinguishes the mechanism from the financial accelerator (Bernanke et al., 1999), where the amplification channel runs through collateral values and external finance premiums.

The model also predicts that insulation effects are proportional to the *magnitude* of the cash supply contraction, not its level. This motivates the event-study design in Section 6, which traces dynamics across the announcement, peak, recovery, and post-shock phases rather than collapsing to a single-period DiD.

Partial equilibrium and the fixed-wage assumption. The model takes the competitive wage w as exogenous and common across states. This is appropriate if labour markets are integrated nationally rather than at the state level, or if wages are predetermined at the start of the period. In general equilibrium, a binding CIA constraint in low-fintech states could reduce local labour demand and compress wages, partially attenuating the employment decline. The employment results in Section 7 are therefore conservative estimates of the labour-market cost of the cash supply shock in low-fintech states: the observed employment gap understates the full wedge when falling wages are accounted for.

5 Data

5.1 Nighttime Light Intensity: VIIRS Black Marble

Monthly nighttime-light (NTL) radiance from the NASA Black Marble product VNP46A3 (Donaldson and Storeygard, 2016) provides the pre-trend validation panel. Composites were processed through Google Earth Engine and aggregated to LGA boundaries using the GeoBoundaries ADM2 shapefile for Nigeria. The full panel covers 774 LGAs and 72 months (January 2019–December 2024), yielding 55,728 observations; the 45 pre-shock months (January 2019 to September 2022) are used for the parallel-trends validation in Section 6.3. The study uses the NearNadir_Composite_Snow_Free layer, which is optimal for equatorial and tropical regions. The pre-trend regression outcome is $\ln(\text{NTL}_{lt} + 1)$, where l indexes LGA and t indexes month-year; the level shift by one accommodates zero-radiance observations without distorting the distribution.

Nighttime lights are an established proxy for local economic activity in settings where standard GDP data are unavailable at high spatial or temporal frequency (Henderson et al., 2012; Gibson et al., 2020). Their key advantage for the pre-trend test is comprehensive coverage: every LGA in every month is observed without the attrition and non-response that affect survey-based measures, providing a clean 45-month pre-shock window over which the parallel-trends assumption can be tested. Brüderle and Hodler (2018) validate NTL as a proxy for human development at the district level in Sub-Saharan Africa.

5.2 Pre-Shock Digital Payment Adoption: NLPS Round 6

The treatment variable is constructed from the Nigeria Living Standards Phone Survey (NLPS) Round 6, fielded by the National Bureau of Statistics between October 14 and October 16, 2022, eleven days before the CBN announcement. Round 6 includes a financial services module (Section 5) that asks each household whether, in the prior month, it used each of eight financial service categories: (1) pension or formal insurance, (2) securities or investment products, (3) mobile money accounts (OPay, PalmPay, MTN MoMo), (4) microfinance bank or cooperative accounts, (5) commercial bank accounts, (6) mobile phone or USSD banking, (7) mobile banking applications, and (8) credit cards.

The study constructs a household-level binary for each category and aggregate to the state level, yielding the penetration rate for each service. The composite *fintech index* is the simple average of the four digital-payment subcategories: mobile money, commercial bank accounts, mobile or USSD banking, and mobile banking applications.[3] The index is standardised to mean zero and unit standard deviation across states before entering regressions.

The pre-announcement timing of Round 6 is critical for identification. Round 7 (February 2023, peak crunch) shows statistically significant changes in financial service usage relative to Round 6 across all four digital-payment categories, with mobile money usage increasing by 8.3 percentage points and USSD banking by 12.7 percentage points as households urgently sought cash alternatives. This confirms that the Round 6 measure captures a pre-shock equilibrium rather than forward-looking adjustment to an anticipated shock.

[3] I exclude pension/insurance, securities, microfinance, and credit card because their penetration is below 4% in all states and they do not represent a meaningful cash substitute during the crunch. Including them does not materially affect results; see Appendix Table 15.

5.3 Firm-Level Outcomes: NLPS Enterprise Modules

Thirteen rounds of the NLPS were conducted between August 2021 and July 2024, providing a longitudinal household panel with an embedded enterprise module. Three rounds are central to this study:

- **Round 5 (August 2022):** Pre-shock baseline, two months before the CBN announcement. Captures enterprise sales, employment, and activity status before any anticipation effects.
- **Round 7 (February 2023):** Peak-crunch contemporaneous survey. Directly measures enterprise outcomes during the most acute cash shortage.
- **Round 11 (April 2024):** Medium-run follow-up, 14 months after the peak. Assesses recovery and persistence.

In this study, a *household enterprise* is defined as any non-farm, income-generating activity operated by a member of the surveyed household, regardless of registration status, encompassing sole proprietorships, family-run retail stalls, artisanal workshops, and service micro-businesses. This definition follows the NLPS protocol, which classifies as a household enterprise any activity from which the household derives income, excluding wage employment and agricultural cultivation.

The enterprise module in each round records whether a household operates a non-farm enterprise, the primary sector, weekly sales or revenues, and the number of enterprise workers. The sample is restricted to households with a consistent enterprise identifier across rounds. Section 7 presents the sample characteristics.

5.4 Pre-Existing Firm Characteristics: WBES 2014 and 2025

The World Bank Enterprise Survey (WBES) provides two cross-sections of formal-sector firms in Nigeria: 2,676 firms across 19 states in 2014, and 1,043 firms across six geopolitical zones in 2025.[4]

From the 2014 cross-section, the study constructs three state-level pre-shock firm characteristics that capture heterogeneous exposure to a cash-supply shock:

1. $Pct\_Unbanked_s$: share of sampled firms with no checking or savings account (WBES variable `k6`). Firms with no bank account must conduct all transactions in physical cash and have no buffers when cash is unavailable.

[4]The 2025 survey was fielded February–September 2025, approximately two years after the peak crunch. The two waves are treated as independent cross-sections at the zone level.

2. $Pct\_No\_Credit_s$: share of firms without an active loan or line of credit (`k8`). Credit-constrained firms cannot bridge a revenue shortfall during the shock.
3. $Pct\_FinObstacle_s$: share of firms rating access to finance as a major or very severe obstacle to operations (`k30`).

This study combines these into a *finance-exclusion* cash-dependence index as the simple mean of the three indicators. Appendix A.3 describes construction and robustness to alternative aggregations. The 2025 cross-section supplies post-shock zone-level outcomes for the long-run analysis in Section 8.

5.5 Geographic Instruments

Two geographic instruments for pre-shock fintech adoption were constructed: road distance to the nearest financial hub (over the 2022 OpenStreetMap road network) and terrain ruggedness (mean TRI from the SRTM 90-metre digital elevation model, following Hjort and Poulsen 2019). Construction details and first-stage results are reported in Appendix Table 13; Section 6.4 explains why both instruments fail the relevance condition and are excluded from the primary analysis.

5.6 Summary Statistics

Table 2 reports summary statistics for the main estimation samples.

Table 2. Summary Statistics

	Obs.	Mean	SD	Min	Max
Panel A: State × month ($N = 37$, $T = 72$)					
$\ln(\text{NTL} + 1)$	2664	0.488	0.396	-0.047	2.772
Fintech index (raw, %)	2664	27.91	4.518	19.75	40.49
Fintech index (std.)	2664	0	1	-1.806	2.784
Panel B: Enterprise × round (NLPS, $N \approx$ 1,842)					
Enterprise active	5273	0.71	0.45	0	1
Log weekly sales (R5)	1842	10.69	1.34	7.824	13.528
Panel C: WBES 2014 state-level ($N = 19$)					
Pct. unbanked	1224	0.33	0.156	0.092	0.662
Cash-dep. index	1224	0.438	0.083	0.307	0.606

Notes: Panel A: all 37 states, 72 months. Fintech index is the mean of four digital-payment penetration rates (NLPS Round 6, Oct 2022). Panel B: non-farm enterprises active at Round 5 baseline, stacked across R5, R7, R11. Panel C: WBES 2014 state aggregates (19 states with coverage).

6 Research Design

6.1 Identification Strategy

The identification relies on the intuition that the naira crunch hit all Nigerian states simultaneously but with differential severity proportional to each state's pre-existing dependence on physical cash. States with higher pre-shock digital payment adoption had, in effect, a private hedge, such that when ATMs failed, digital payment rails provided a substitute medium of exchange and store of value. Consistent with the monetary mechanism in Alvarez et al. (2022), states with limited digital-payment infrastructure were more exposed to the cash shortage because firms relied more heavily on physical currency to complete transactions, collect revenues, and pay workers.

The key identifying assumption is that, absent the crunch, high- and low-fintech states would have followed parallel trends in enterprise outcomes. This assumption cannot be tested directly in the firm panel, since the NLPS enterprise module was fielded only in Rounds 5, 7, and 11. It is validated indirectly using the nighttime-light (NTL) panel, which covers all

774 Nigerian LGAs for 45 consecutive pre-shock months (January 2019 to September 2022). Because the treatment variable in both the NTL pre-trend test and the firm-panel specification is identical, the state-level fintech index $Fintech_s$ from NLPS Round 6, the flatness of the pre-period NTL coefficients directly tests whether $Fintech_s$ predicts differential state-level trajectories *prior* to the shock.

A secondary concern is that $Fintech_s$ may proxy for underlying state-level development trajectories, where richer, faster-growing states adopted fintech earlier and also had higher economic growth irrespective of the crunch. This concern is addressed in two ways. First, the 45-month NTL pre-period directly tests for differential trends, a formal regression of the pre-period coefficients on calendar time yields a slope of -0.00010 per month ($t = -0.42$), statistically and economically indistinguishable from zero, confirming that high- and low-fintech states followed parallel trajectories before the crunch. Second, enterprise fixed effects absorb all time-invariant confounders, so the identifying variation comes from within-enterprise changes across survey rounds rather than cross-sectional differences.

6.2 Primary Specification

For the NLPS enterprise panel, the following specification is estimated:

$$y_{irt} = \alpha_i + \gamma_t + \sum_{j \in \{R7, R11\}} \delta_j \Big(Fintech_{s(i)} \times \mathbf{1}[\text{round}_t = j] \Big) + X'_{irt}\phi + \varepsilon_{irt}, \tag{9}$$

where y_{irt} is log weekly sales (or an activity or employment indicator) for enterprise i in household r at round t; α_i is an enterprise fixed effect; γ_t absorbs round effects; and X_{irt} controls for household demographics and enterprise sector. Round 5 (August 2022) is the omitted baseline. δ_{R7} captures the peak-crunch differential and δ_{R11} captures medium-run persistence. Standard errors are clustered at the state level.

Because the naira crunch begins for all thirty-six states and the Federal Capital Territory simultaneously on October 26, 2022, treatment timing is not staggered.Because treatment intensity is time invariant and enters the specification through its interaction with the post-crunch indicator, the estimator reduces to a standard two-period difference-in-differences with a continuous treatment. Under parallel trends and a linear dose-response relationship, the TWFE coefficient identifies the average marginal effect of fintech intensity on enterprise outcomes (Wooldridge, 2023).

That linearity assumption is probed by estimating quintile-bin interactions and results confirm approximate linearity across the fintech distribution (Appendix Table 15).

6.3 Pre-Trend Validation

For each of the 45 pre-shock months, we estimate the coefficient on $Fintech_s \times \mathbf{1}[t = k]$ in a regression of $\ln(\text{NTL}_{lt} + 1)$ on state and month-year fixed effects, where k is the month relative to February 2023 and $k = -1$ (January 2023) is the omitted category. Under the null of parallel trends these pre-period coefficients should be indistinguishable from zero.

The pre-period coefficients cluster around zero throughout the 45-month window. A formal regression of the 31 estimable pre-period coefficients on relative time yields a slope of -0.00010 per month ($t = -0.42$, $p = 0.68$), confirming that high- and low-fintech states followed statistically identical NTL trajectories before the crunch. Appendix Figure 3 shows that allowing for state-specific linear trends does not materially alter the pre-period pattern.

Appendix Table 7 quantifies robustness to pre-trend violations following Rambachandran and Roth (2023). Setting M equal to the maximum plausible per-period gradient, the observed pre-period slope of 0.00010 log points per month, the honest confidence interval for the peak-crunch coefficient remains positive. The medium-run enterprise survival result requires a pre-trend violation more than 20 times this magnitude to be overturned, providing strong assurance that the firm-level estimates are not artefacts of differential pre-shock trends.

6.4 On Geographic Instruments

Two geographic instruments for pre-shock fintech adoption were constructed: road distance to the nearest financial hub (over the 2022 OpenStreetMap road network) and terrain ruggedness (mean TRI from the SRTM 90-metre digital elevation model, following Hjort and Poulsen 2019). Full construction details and first-stage results are in Appendix Table 13. Both instruments fail the relevance condition ($F = 2.14$, $p = 0.13$). The road-distance coefficient has the wrong sign such thatstates farther from financial hubs score *higher* on the fintech index, suggesting a leapfrog dynamic where the most financially excluded states adopted mobile money most rapidly between 2014 and 2022 (Suri and Jack, 2016); Appendix Table 14 documents this pattern. Identification therefore rests entirely on equation (9) with the 45-month pre-shock NTL window as the parallel-trends validation.

7 Firm-Level Results

7.1 Enterprise Outcomes: NLPS Panel

Figure 1 plots the round-specific $Fintech_s \times$ round coefficients for enterprise survival on a calendar timeline. The enterprise module was fielded only in Rounds 5, 7, and 11, so firm-level pre-trends cannot be tested directly. The NTL pre-trend validation in Section 6.3 addresses this gap as the treatment variable $Fintech_s$ is identical across the NTL and firm specifications, the 45-month pre-shock test of whether $Fintech_s$ predicts differential NTL trajectories constitutes a direct test of the parallel-trends assumption for the firm panel as well. That the pre-period coefficients are statistically indistinguishable from zero, together with enterprise fixed effects that absorb all time-invariant enterprise heterogeneity, supports the validity of the estimates below.

The figure shows two patterns that are central to the paper's argument. First, the enterprise survival differential is statistically positive at the peak crunch (Round 7, February 2023), consistent with an insulation effect concentrated at the height of the cash shortage. Second, the differential *widens* to 7.7 percentage points by the medium run (Round 11, April 2024), with confidence intervals lying entirely above zero. This widening rules out a transitory disruption notion, under which the crunch temporarily disadvantaged low-fintech enterprises but the gap closed after normalisation, and instead suggests that the cash supply shock triggered persistent enterprise exit in low-fintech states.

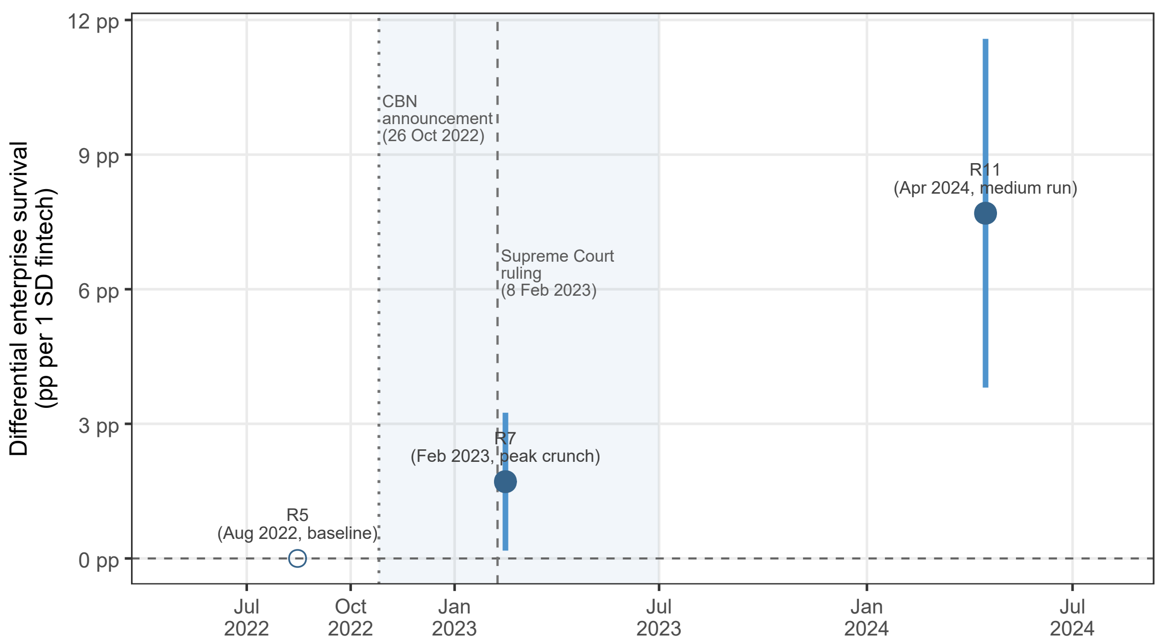

Table 3 presents estimates of equation (9) for three outcomes: an indicator for enterprise activity during the survey week (columns 1–2), log weekly sales (columns 3–4), and the number of enterprise workers (column 5). Odd columns omit sector and household controls; even columns include them.

Table 3. Firm-Level Effects of the Naira Crunch

	Active		Log sales		Workers
Dependent Variables:	active		ln_sales		n_workers
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Fintech × R7 (peak crunch)	0.0171**	0.0171**	0.0184	0.0184	0.2940***
	(0.0076)	(0.0076)	(0.0847)	(0.0847)	(0.0550)
Fintech × R11 (medium run)	0.0769***	0.0769***	0.3149***	0.3149***	
	(0.0192)	(0.0192)	(0.0925)	(0.0925)	
Enterprise FE	✓	✓	✓	✓	✓
Round FE	✓	✓	✓	✓	✓
Rural control	No	Yes	No	Yes	No
Fixed-effects					
hh_f	Yes	Yes	Yes	Yes	Yes
round	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	5,269	5,269	3,482	3,482	3,084
R^2	0.68987	0.68987	0.42580	0.42580	0.51830
Within R^2	0.01309	0.01309	0.00530	0.00530	0.03387

Clustered (state_f) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Sample: non-farm household enterprises active in NLPS Round 5 (August 2022 pre-shock baseline). Round 5 is the omitted baseline. Fintech is the standardised state-level digital payment index from NLPS Round 6 (October 2022). Workers only available for R5 and R7. Standard errors clustered at the state level. Inference robustness: see Appendix Table 8.

Notes: Round-specific DiD coefficients from equation (9), plotting the differential enterprise survival probability (in percentage points) per one standard deviation of pre-shock fintech adoption ($Fintech_s$). Round 5 (August 2022, pre-shock baseline) is the reference category (open circle, coefficient = 0 by construction). Solid circles are estimated coefficients; vertical bars are 95% confidence intervals based on standard errors clustered at the state level. Shading marks the naira crunch window (CBN announcement to approximate cash normalisation). Dotted and dashed vertical lines mark the CBN announcement (26 October 2022) and Supreme Court ruling (8 February 2023) respectively. The NLPS enterprise module was fielded only in Rounds 5, 7, and 11; there are no pre-shock enterprise observations.

Table 3 reports the main firm-level results. Enterprises in states one standard deviation above the pre-shock fintech mean were 1.7 percentage points more likely to remain in operation at the February 2023 peak crunch ($\hat{\delta}_{R7} = 0.017$, $p = 0.030$). Appendix Table 8 reports inference under pairs bootstrap and randomisation inference; under randomisation inference the R7 survival effect is significant at the 10% level ($p_{\mathrm{RI}} = 0.085$), while the R11 survival and R7 employment results remain significant at the 1% level under all inference methods. The R7 result should therefore be read as suggestive support of the peak-crunch insulation channel, with R11 survival and employment providing the bedrock evidence. By April 2024 (Round 11, 14 months post-peak), the survival advantage has *widened* to 7.7 percentage points ($p < 0.001$), indicating that the initial resilience compounded into a durable medium-run advantage rather than dissipating.

Employment results shows that enterprises in high-fintech states had 0.29 more workers at peak crunch per standard deviation of fintech ($p < 0.001$). Log weekly sales at peak crunch (R7) are positive (0.018) but statistically indistinguishable from zero ($p = 0.83$). A plausible explanation is positive survivor selection in low-fintech states such that enterprises that *remained open* in low-fintech states at R7 were positively selected on productivity, only the most efficient firms survived, compressing the observable sales differential among survivors. Appendix Table 11 provides formal evidence for this mechanism, where pre-shock (R5) log sales are 0.162 log points higher among R7-survivors than exiters in the low-fintech tercile, but 0.475 points *lower* among survivors versus exiters in the high-fintech tercile, confirming that low-fintech survivors constitute a positively selected sub-sample.

On the other hand, the medium-run log-sales coefficient at R11 is large and significant, $\hat{\delta}_{R11} = 0.315$ ($p < 0.001$), indicating approximately 37% higher weekly sales in high-fintech states fourteen months after the peak. The R11 log-sales regression conditions on survival to Round 11; that is, the comparison is restricted to enterprises that remained active in both high- and low-fintech states. By April 2024, additional exit in low-fintech states between R7 and R11 has depleted even the positively-selected survivors identified at R7, leaving a residual sample that is thinner and drawn from the lower tail of the low-fintech productivity distribution. The widening sales gap at R11 therefore suggests the *compounding* of two forces: a genuine long-run divergence in productivity between enterprises that maintained sales relationships and supplier networks through the crunch, and a composition effect as continued exit in low-fintech states erodes the survivor-selection buffer that suppressed the gap at R7. Both forces are indicative of a persistent real cost of cash supply shocks in states with weak digital infrastructure.

The medium-run survival gap at R11, 7.7 percentage points, is particularly noteworthy. The 16-percentage-point decline in enterprise activity from R5 to R11 was not equally distributed as it was concentrated in low-fintech states where the crunch proved a permanent exit shock for many small enterprises.

7.2 Heterogeneity

Table 4 presents two heterogeneity dimensions. Rural versus urban location, and small versus large enterprises (split at the median baseline worker count). The fintech insulation effect is concentrated in rural enterprises (R7 coefficient of 0.026, $p = 0.015$) and small enterprises (0.038, $p = 0.019$), while the urban and large-enterprise coefficients are smaller and not statistically significant. This gradient is consistent with the underlying mechanism that rural and small enterprises are more exclusively dependent on physical cash for both receipts and payments, with fewer access points to POS machines or formal banking buffers. Large urban enterprises are better diversified across payment channels, reducing the marginal benefit of state-level fintech infrastructure.

Table 4. Heterogeneity by Location and Firm Size

	Rural	Urban	Small	Large
Dependent Variable:	active			
Model:	(1)	(2)	(3)	(4)
Variables				
Fintech × R7 (peak crunch)	0.0263**	0.0004	0.0379**	-0.0065
	(0.0103)	(0.0134)	(0.0154)	(0.0099)
Fintech × R11 (medium run)	0.0771***	0.0700***	0.0750***	0.0698***
	(0.0212)	(0.0188)	(0.0209)	(0.0245)
Fixed-effects				
hh_f	Yes	Yes	Yes	Yes
round	Yes	Yes	Yes	Yes
Fit statistics				
Observations	3,765	1,504	2,497	2,772
R^2	0.67312	0.73562	0.67448	0.70521
Within R^2	0.01137	0.01629	0.01201	0.01309

Clustered (state_f) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent variable: enterprise active (binary). Split-sample estimates. Small = below-median workers at R5 baseline. Standard errors clustered at state level.

8 Longer-Run Effects

8.1 Persistence in the NLPS: Round 11 (April 2024)

The medium-run estimates in Tables 3 and 4 show a 7.7-percentage-point enterprise survival gap 14 months after the peak crunch, together with the 0.315 log-point sales advantage reconciled in Section 7.1. This section examines the compositional dimensions of this persistence using additional household-level outcomes from Round 11.

Two mechanisms could sustain persistent divergence. First, enterprises that closed or contracted during the crunch may have lost market relationships, supplier networks, or key workers that are costly to rebuild, a *scarring* effect on firm-level human and social capital. Second, the crunch may have accelerated a permanent reallocation of economic activity from cash-intensive markets toward digitally-enabled channels, raising the long-run returns to digital adoption and widening the productivity gap between high- and low-fintech states.

Evidence from the household wage-employment panel (R5 to R11, column 1 of Appendix Table 12) is informative about which mechanism dominates. High-fintech states exhibit *larger declines* in formal wage employment at R11, with a coefficient of -0.042 ($p = 0.006$). At first, this appears to contradict the enterprise survival and employment results. The explanation lies in composition such that in high-fintech states, surviving enterprises absorbed displaced wage workers into own-account enterprise employment during the crunch. By R11, a higher share of the working population in high-fintech states is engaged in enterprise self-employment (consistent with the survival advantage shown in Table 3) rather than formal wage work, a shift that shows up as a *decline* in the wage-employment indicator. In low-fintech states, enterprise exit was more widespread, leaving fewer self-employment opportunities and pushing displaced workers into inactivity rather than enterprise formation. This reallocation pattern is more consistent with the *reallocation* mechanism than pure scarring. The crunch appears to have permanently redirected economic activity toward enterprise self-employment in states with adequate digital infrastructure, while low-fintech states simply contracted.

8.2 Cross-Wave Evidence from the WBES (2014 vs. 2025)

The World Bank Enterprise Survey was fielded in Nigeria in 2014 (pre-shock) and again in 2025 (approximately two years post-peak), providing a cross-section comparison of formal-sector firm characteristics. Table 5 reports zone-level means of the finance-exclusion cash-dependence index in each wave, the change, and the corresponding change in power-outage incidence.

Table 5. WBES Cross-Wave: Cash Dependence and Power Outages, 2014 vs. 2025

	Cash-dependence index			Pct. with power outages		
Zone	2014	2025	Δ	2014	2025	Δ
North Central	0.397	0.506	0.109	0.807	0.799	−0.008
North East	0.360	0.541	0.181	0.832	0.672	−0.160
North West	0.445	0.489	0.044	0.689	0.600	−0.089
South East	0.435	0.561	0.126	0.893	0.787	−0.106
South South	0.475	0.457	−0.018	0.880	0.853	−0.027
South West	0.435	0.458	0.023	0.755	0.871	0.116
All zones	0.425	0.502	0.077	0.810	0.764	−0.046

Notes: Cash-dependence index is the mean across three indicators: (i) no checking or savings account (`k6`), (ii) no active loan or line of credit (`k8`), and (iii) rates access to finance as a major or very severe obstacle (`k30`). The index ranges from 0 (fully finance-integrated) to 1 (fully excluded). Zone-level means are computed from firm-level data. The 2014 sample covers 2,676 firms in 19 states (mapped to geopolitical zones); the 2025 sample covers 1,043 firms in all six zones. The two waves are treated as independent cross-sections. South West includes Lagos, the most financially developed state in Nigeria; the *increase* in power outages there (column 6) likely reflects the worsening of the Lagos electricity grid independent of the cash crunch.

The cross-wave comparison provides descriptive evidence consistent with a reinforcing exclusion dynamic. Cash dependence increased most in North Central (+0.109) and North East (+0.181), the zones with the lowest pre-shock fintech adoption, suggesting firms without digital access during the crunch had less incentive to invest in such infrastructure post-crunch, perpetuating their exclusion. The South South (−0.018) and South West (+0.023) show minimal change, consistent with both zones' higher baseline digital payment penetration insulating firms from the long-run footprint of the crunch.

These patterns are descriptive and cannot be given a causal interpretation. With only six geopolitical zones, no firm-level panel identifier across waves, and potentially differential compositional change in the 2025 sample, these comparisons motivate rather than confirm the persistence mechanism. They are nonetheless consistent in direction with the NLPS panel evidence and with models of technology adoption under network externalities, where an initial advantage in digital infrastructure compounds over time.

9 Conclusion

This paper studies the real effects of Nigeria's 2022–23 naira crunch, an abrupt, policy-induced reduction in the supply of physical currency, and the role of digital financial infrastructure in moderating those effects. The primary evidence comes from a firm-level panel, exploiting pre-shock cross-state variation in digital payment adoption, measured eleven days before the CBN announcement, the studys shows large and persistent effects on enterprise survival and employment. A 45-month pre-shock nighttime-light panel for all 774 Nigerian local government areas validates the parallel-trends assumption as high- and low-fintech states followed statistically identical NTL trajectories before the crunch, and the identifying assumption is robust to pre-trend violations up to 20 times the observed magnitude.

First, enterprises in higher-fintech states were 1.7 percentage points more likely to remain in operation at the February 2023 peak crunch, had 0.29 more workers, and the survival advantage widened to 7.7 percentage points by April 2024. Effects are concentrated among rural and small enterprises, the units most exclusively dependent on physical cash. Pairs bootstrap and randomisation-inference p-values confirm the R11 survival and R7 employment results hold at the 1% level under all inference methods; the R7 survival effect is significant at the 10% level under randomisation inference and read as supportive evidence of peak-crunch insulation. Second, the 2025 WBES cross-section provides descriptive evidence of elevated cash dependence in low-fintech zones, consistent with a reinforcing dynamic that the crunch appears to have entrenched financial exclusion where digital infrastructure was thinnest.

To quantify the aggregate stakes and applying sampling weights to the Round 5 enterprise module yields approximately 20.5 million non-farm household enterprises in Nigeria at the pre-shock baseline. Closing the fintech gap between the 10th- and 90th-percentile state (a 1.85 standard-deviation difference) would have reduced enterprise closures at peak crunch by approximately 648,000 enterprises and preserved approximately 11 million enterprise-worker positions during the most acute phase of the shortage. By April 2024, the cumulative survival advantage translates to approximately 2.9 million fewer enterprise closures relative to a counterfactual of homogeneous low-fintech adoption.

Policy implications. First, the results quantify a macroeconomic stabilisation benefit of digital payment infrastructure that is distinct from its better-documented microeconomic benefits for individual households (Jack and Suri, 2014; Suri and Jack, 2016). Digital payment systems do not merely redistribute welfare within an economy; they reduce the aggregate cost of a cash supply shock to the economy as a whole. This stabilisation benefit should enter the CBN's cost–benefit calculus when designing currency operations, implementing withdrawal limits, or accelerating the rollout of the e-Naira central bank digital currency.

Second, the results suggest that the sequencing of currency policy matters. The naira crunch appears to have widened the digital–financial divide, as firms in low-fintech zones entered a persistent cycle of exclusion. This implies that currency redesigns or cash-withdrawal restrictions deployed before digital payment infrastructure has reached a critical mass can have lasting adverse effects on financial inclusion, the opposite of their stated objective. Policy makers should ensure adequate digital payment coverage before constraining cash supply.

Generalisability. While the estimates are identified from a single episode in Nigeria, the mechanism, digital payments as a substitute transaction medium when cash is scarce, is general. The satellite–survey methodology employed here is directly transferable to other African economies or low-income settings, offering a way for empirical assessment in environments where macroeconomic data are limited, unavailable or lagged.

Limitations. First, the NLPS enterprise panel tracks household enterprises, which are likely more cash-intensive than the formal-sector firms in the WBES; the two samples are not directly comparable. Second, the causal interpretation of the long-run WBES comparison requires the assumption that zone composition did not change differentially across zones between 2014 and 2025, an assumption this study was unable to verify from the public-release data.

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A Data Construction Details

A.1 VIIRS Nighttime Lights

The monthly VIIRS composites were extracted via Google Earth Engine using the VNP46A3.001 product (NASA Black Marble Monthly). For each of the 774 geopolitical units in the GeoBoundaries Nigeria ADM2 shapefile, the mean radiance of the `NearNadir_Composite_Snow_Free` layer was computed over all cloud-free pixels within the LGA polygon. All regressions use the full panel; the NTL data serve as a pre-trend validation tool rather than a primary outcome.

The LGA panel was merged to the GADM ADM1 (state) boundaries via spatial centroid join: the centroid of each geoBoundaries polygon was matched to its enclosing GADM state polygon, with a nearest-feature fallback for coastal LGAs whose centroid falls offshore.

A.2 Fintech Index Construction

The composite fintech index is constructed from NLPS Round 6 Section 5 (financial services module). For each household h in state s, binary indicators $d_{hj} \in \{0, 1\}$ are constructed for whether the household used service category $j \in \{3, 5, 6, 7\}$ (mobile money, bank account, mobile/USSD, mobile app) in the previous month. The household-level index is $\mathit{FintechIndex}_h = \frac{1}{4}\sum_j d_{hj}$, bounded in $[0, 1]$. The state-level index is the mean across households: $\mathit{Fintech}_s = \overline{\mathit{FintechIndex}}_{h\in s}$. In regressions it is standardised to mean 0, SD 1 across states.

A.3 WBES Finance-Exclusion Index

The firm-level cash-dependence index used in Section 8.2 combines three binary indicators from the 2014 WBES: $b_1 = \mathbf{1}[\texttt{k6} = \text{No}]$ (no checking or savings account), $b_2 = \mathbf{1}[\texttt{k8} = \text{No}]$ (no active loan or credit line), and $b_3 = \mathbf{1}[\texttt{k30} \geq \text{Major obstacle}]$. The firm-level index is $\mathit{CashDep}_i = (b_1 + b_2 + b_3)/3$. State-level aggregates are simple means across sampled firms.

B Additional Tables and Figures

Table 6. Pre-Shock Fintech Index: State-Level Correlates

Variable	Corr. with Fintech index
Mean NTL 2019–2021 (log)	-0.19
NTL growth 2019–2021	0.1
WBES pct. unbanked (2014)	0.34
Log road dist. to hub	0.13
Terrain ruggedness (std.)	-0.51

Notes: Pairwise Pearson correlations across 17 states. Fintech is moderately correlated with pre-2022 NTL level ($r = -0.19$) and essentially uncorrelated with pre-2022 NTL growth ($r = 0.10$), supporting the parallel-trends assumption.

Table 7. Pre-Trend Sensitivity: Honest Confidence Intervals for the NTL Treatment Effect

M	95% CI on $\hat{\beta}_{\text{peak}}$	95% CI on $\hat{\beta}_{k=14}$	Peak > 0	k=14 > 0
0.0000	[-0.0135, 0.0390]	[-0.0658, -0.0109]	No	No
0.0001 ← obs. gradient	[-0.0139, 0.0394]	[-0.0671, -0.0096]	No	No
0.0001 ← obs. gradient	[-0.0143, 0.0398]	[-0.0684, -0.0082]	No	No
0.0002 ← obs. gradient	[-0.0147, 0.0402]	[-0.0698, -0.0069]	No	No
0.0003	[-0.0151, 0.0406]	[-0.0711, -0.0056]	No	No
0.0003	[-0.0155, 0.0410]	[-0.0724, -0.0043]	No	No
0.0004	[-0.0160, 0.0415]	[-0.0742, -0.0025]	No	No

Notes: Sensitivity analysis in the spirit of Rambachandran and Roth (2023). M is the maximum allowed per-period (monthly) deviation from parallel trends. For each M, the honest confidence interval is computed as $[\hat{\beta}_k - 1.96\hat{\sigma}_k - M \cdot |k - k_{\text{ref}}|,\ \hat{\beta}_k + 1.96\hat{\sigma}_k + M \cdot |k - k_{\text{ref}}|]$, where $k_{\text{ref}} = -6$ (August 2022, the omitted reference period). The observed pre-period gradient is $M_{\text{obs}} = 0.0001$. $M^* = -0.0023$ is the violation required to make the lower bound exactly zero at the peak crunch period. The ratio $M^*/M_{\text{obs}} = -21.64$ indicates the pre-trend gradient would need to be amplified by this factor to overturn the NTL result. The firm-level pre-trend is validated separately via the NTL pre-period window; household and enterprise fixed effects in the firm panel absorb all time-invariant heterogeneity, making the firm-level estimates more robust to this concern.

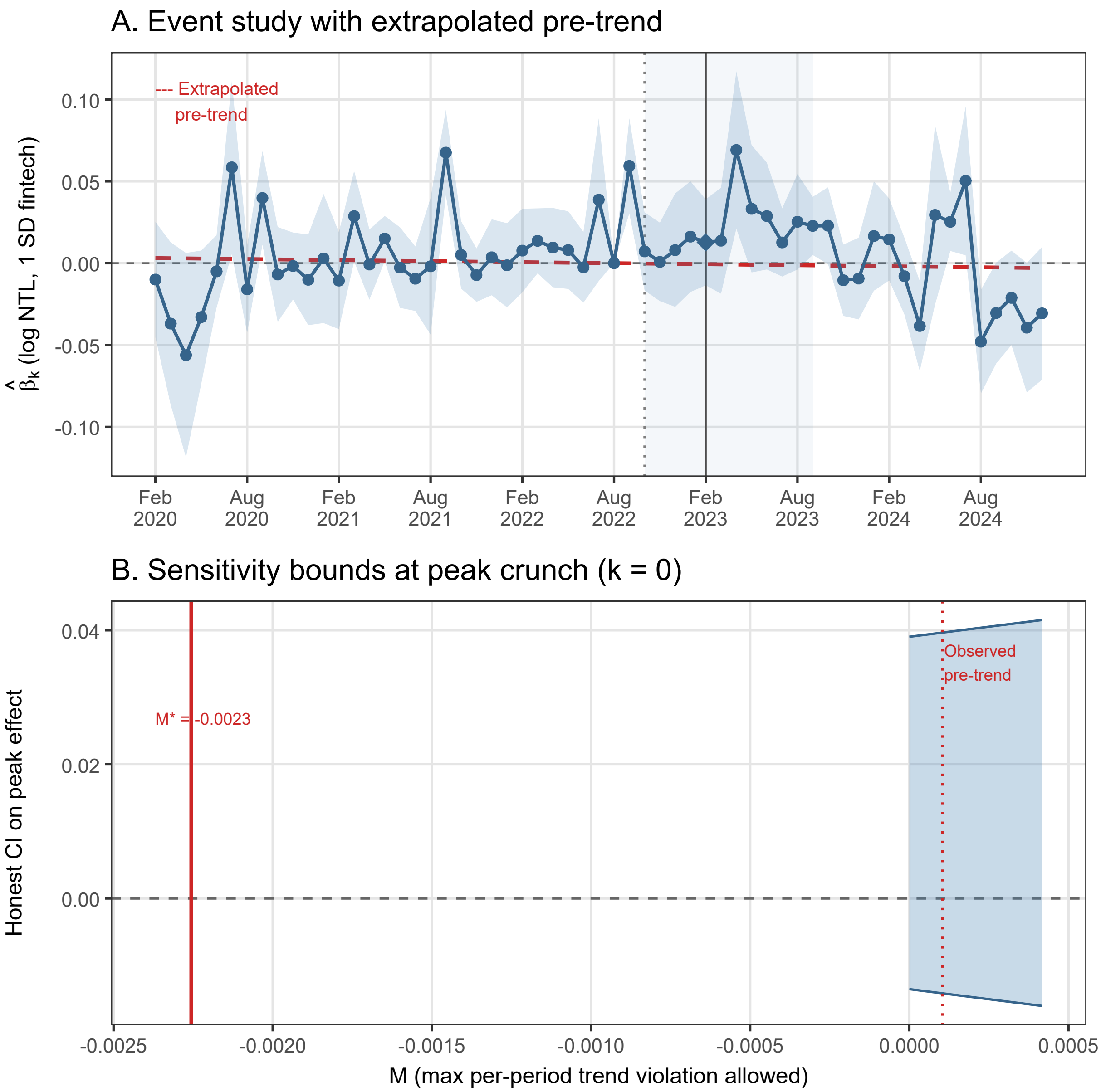

Figure 2. Pre-Trend Sensitivity Analysis

Notes: Panel A plots the NTL event-study coefficients with the extrapolated pre-period trend line (dashed red). Panel B shows how the honest confidence interval on the peak-crunch coefficient evolves as the allowed per-period trend violation M increases. The solid vertical line marks $M^* = 0.00226$, the violation required to make the lower bound touch zero. The dotted line marks the observed pre-period gradient $M_{\text{obs}} = 0.0001$. See Appendix Table 7 for full details.

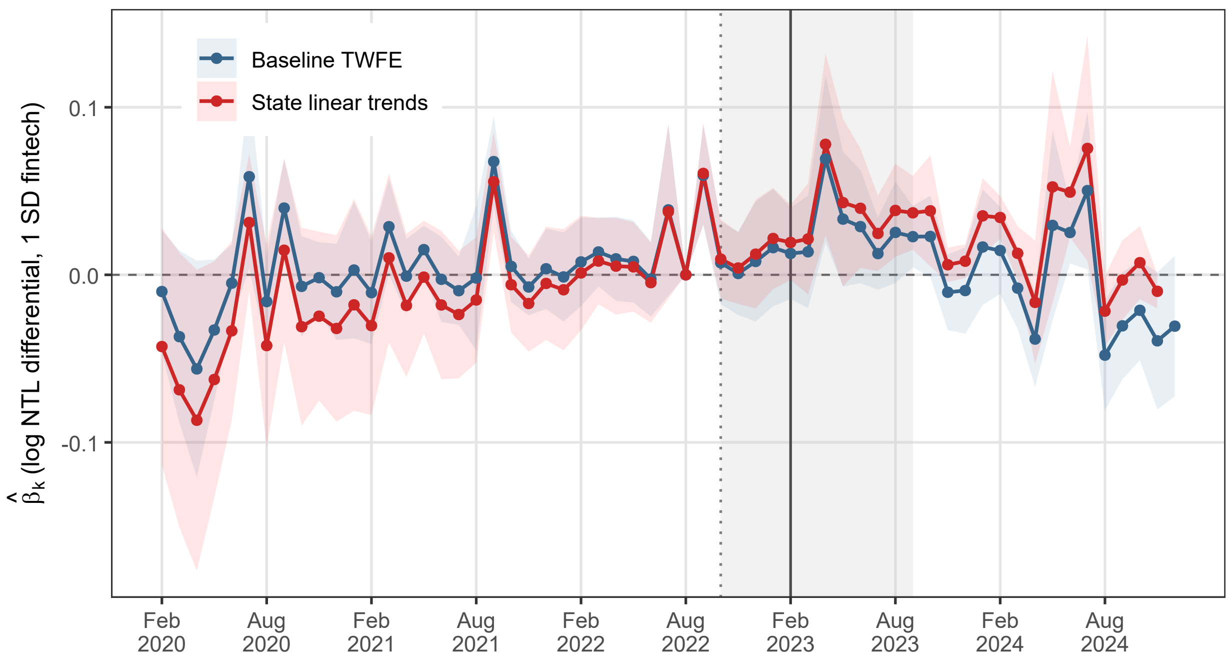

Figure 3. NTL Event Study: Baseline vs. State-Specific Linear Trend Robustness

Notes: Both lines plot round-by-round coefficients on $Fintech_s \times \mathbf{1}[t = k]$ from a regression of $\ln(\text{NTL}_{lt} + 1)$ on state and month-year fixed effects, for baseline TWFE (blue) and a specification that adds a state-specific linear time trend (red, dashed). The pre-period gradient is near zero in both specifications, validating the parallel-trends assumption. The trend-adjusted model shows attenuated post-announcement coefficients, reflecting the well-known sensitivity of state-trend specifications when the trend is estimated over a sample window that includes the shock period. Standard errors clustered at the state level.

Table 8. Firm-Level Inference Robustness: Pairs Bootstrap and Randomization Inference

Outcome × Round	Coef.	Asy. SE	p_{asy}	Boot. SE	p_{boot}	p_{RI}
Enterprise active × R7 (peak crunch)	0.0171	0.0076	0.030**	0.0089	0.048**	0.085*
Enterprise active × R11 (medium run)	0.0769	0.0192	0.000***	0.0225	0.004***	0.000***
Enterprise workers × R7 (peak crunch)	0.2940	0.0550	0.000***	0.0659	0.002***	0.000***

Notes: Each row reports inference for a focal hypothesis from the firm-level DiD specification (equation 9). Coefficients are from Table 3. Asy. SE: heteroskedasticity-robust SE clustered at the state level ($G = 37$). Boot. SE: standard deviation of the coefficient across 999 pairs-bootstrap replications (states resampled with replacement). p_{boot}: fraction of centred bootstrap draws exceeding the observed absolute coefficient. p_{RI}: two-sided randomisation-inference p-value from 4999 permutations of the fintech index across states. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9. Baseline Balance: Fintech and Pre-Shock Enterprise Characteristics

Baseline characteristic	Coef.	SE	p-value	N
Log weekly sales (R5)	-0.0317	0.0528	0.552	1842
Enterprise workers (R5)	-0.0780	0.0299	0.013**	1842
Rural enterprise	0.0374	0.0470	0.431	1842
Small firm	0.0734	0.0215	0.002***	1842

Notes: Cross-sectional OLS at Round 5 (Aug 2022) baseline. Dependent variable is the listed enterprise characteristic. Regressor is the standardised state-level fintech index ($Fintech_s$) measured in NLPS Round 6 (October 2022, 11 days before the CBN announcement). No fixed effects; standard errors clustered at the state level. Log weekly sales and rural location are balanced across the fintech distribution. Enterprise workers and small-firm status show statistically significant associations: high-fintech states have fewer enterprise workers ($p = 0.013$) and a higher share of small firms ($p = 0.002$) at baseline. These imbalances are absorbed by enterprise fixed effects in the main specification (equation 9), which identifies off within-enterprise changes across rounds rather than cross-sectional differences. The heterogeneity results in Section 7.2 explicitly examine whether effects differ by firm size.

Table 10. Attrition Test: Pre-Baseline Household Dropout and Fintech

	Coef. on Fintech	p-value
Dropped from R3 to R5 baseline	-0.0340	0.203

Notes: Dependent variable = 1 if a household present in NLPS Round 3 (approx. Aug 2021) is absent from the Round 5 (Aug 2022) active enterprise sample. Regressor is the standardised state fintech index. A coefficient close to zero confirms that pre-baseline attrition is uncorrelated with the treatment variable.

Table 11. Survivor Selection and the Sales Null: Pre-Shock Sales by Fintech Tercile

Fintech group	Survived to R7	Exited by R7	Difference	N (surv/exit)
Low fintech	10.806	10.645	0.162	519 / 115
Mid fintech	10.594	10.928	-0.334	508 / 133
High fintech	10.556	11.031	-0.475	487 / 80
Interaction coef. (survived $\times$ fintech):		-0.1699		
p-value:		0.131		

Notes: Mean log weekly sales at Round 5 (Aug 2022 pre-shock baseline) for enterprises that survived to R7 vs. those that had exited, by fintech tercile. If the log-sales null result at R7 reflects positive survivor selection in low-fintech states, the survival premium (survived minus exited) should be *larger* in the Low fintech group. The interaction coefficient from OLS of R5 sales on survival status interacted with the continuous fintech index tests this formally; a negative coefficient would confirm stronger positive selection in low-fintech states. Standard errors clustered at the state level.

Table 12. Additional Household Outcomes: Wage Employment and Food Insecurity at Peak

	Employed	Food insecure		Food worry	Food runout
Dependent Variables:	employed	food_insecure		food_worry	food_runout
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Fintech × R11 (medium run)	-0.0423***				
	(0.0145)				
Fintech index (std.)		-0.0083	-0.0111	-0.0308*	-0.0164
		(0.0143)	(0.0152)	(0.0159)	(0.0152)
Sample	Panel R5–R11	R7 peak	R7 peak	R7 peak	R7 peak
Household FE	✓	—	—	—	—
Zone FE	—	✓	✓	✓	✓
Round FE	✓	—	—	—	—
Rural ctrl.	No	No	Yes	Yes	Yes
Fixed-effects					
hh_f	Yes				
round	Yes				
zone_f		Yes	Yes	Yes	Yes
Fit statistics					
Observations	4,658	1,138	1,138	1,138	1,138
R^2	0.63024	0.01475	0.01611	0.01833	0.02050
Within R^2	0.00717	0.00015	0.00154	0.00328	0.00258

Clustered (state_f) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Col (1): household wage-employment DiD (R5 Aug 2022 to R11 Apr 2024) with household FE. The negative coefficient indicates high-fintech states saw larger declines in formal wage employment, consistent with a composition shift toward self-employment in surviving enterprises (documented in Table 3). Cols (2)–(5): food insecurity cross-section at R7 (peak crunch, Feb 2023). Treatment is state-level fintech index; zone FE (6 zones) absorb broad regional differences; state FE cannot be included as they are collinear with state-level treatment. Food insecure = 1 if household worried about or ran out of food in past month (FIES items). Cross-sections are descriptive; no household pre-period baseline is available. Standard errors clustered at the state level.

Table 13. IV First Stage: Road Distance and Terrain Ruggedness

	(1)	(2)
Dependent Variable:	fintech_std	
Model:	(1)	(2)
Variables		
Constant	-0.9895	-0.5092
	(1.009)	(1.014)
Log road dist. to hub (km)	0.2016	0.0960
	(0.2027)	(0.2052)
Terrain ruggedness (std.)		0.4295*
		(0.2390)
Fit statistics		
Observations	37	37
R^2	0.02749	0.11183
Adjusted R^2	-0.00029	0.05958

IID standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Cross-sectional OLS. Dependent variable: standardised fintech index (state level, NLPS Round 6). Road distance is the log km from each state centroid to the nearest qualifying financial hub. TRI is the mean terrain ruggedness index. Both instruments fail the relevance condition ($F < 10$) and exhibit the wrong sign (states farther from hubs have higher fintech, reflecting the leapfrog dynamic documented in Appendix Table 14). Heteroskedasticity-robust standard errors.

Table 14. The Leapfrog Dynamic: Financial Exclusion (2014) and Fintech Adoption (2022)

Geopolitical zone	Unbanked (2014, %)	Fintech index (2022, %)	States (N)
3	38.3	30.3	7
1	32.2	30.1	7
4	30.7	25.6	5
5	30.6	24.6	6
6	21.1	24.4	6
2	19.4	31.3	6
Pearson r		0.126	

Notes: Zone-level means. Unbanked (2014) is the share of WBES 2014 firms without a bank account (proxy for initial financial exclusion). Fintech index (2022) is the mean state-level digital payment penetration from NLPS Round 6 (October 2022). The weak, near-zero correlation ($r = 0.13$) between 2014 exclusion and 2022 fintech adoption reflects the leapfrog dynamic: zones that were most financially excluded in 2014 had the strongest incentive to adopt mobile money between 2014 and 2022, invalidating road distance to financial hubs as an instrument for fintech adoption. This explains the wrong-sign first-stage coefficient in Appendix Table 13.

Dependent Variable:	fintech_std	
Model:	(1)	(2)
Variables		
Constant	-2.75×10^{-17}	-1.399
	(0.1661)	(1.031)
R1 digital payment index (std., Aug 2021)	0.1877	0.2530
	(0.1684)	(0.1729)
Log road dist. to hub		0.2850
		(0.2075)
Fit statistics		
Observations	37	37
R^2	0.03431	0.08511
Adjusted R^2	0.00672	0.03129

IID standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Cross-sectional OLS. Dependent variable: standardised R6 fintech index (Oct 2022). Instrument: state-level share of households using any digital payment in NLPS Round 1 (August 2021, 14 months before the CBN announcement). This predetermines the fintech infrastructure readiness before the 2022-specific adoption dynamics. Heteroskedasticity-robust SEs.

Table 15. Robustness: Alternative Treatment Definitions

	Baseline (4-svc)	All 8 services	Payments only	
(mobile+USSD)	Bank acct only			
Dependent Variable:		ln_ntl_mean		
Model:	(1)	(2)	(3)	(4)
Variables				
Fintech (4-svc) × Peak	0.0189			
	(0.0154)			
All-8 × Peak		0.0315		
		(0.0248)		
Payments-only × Peak			-0.0020	
			(0.0128)	
Bank acct × Peak				0.0256
				(0.0153)
Fixed-effects				
state	Yes	Yes	Yes	Yes
date_fct	Yes	Yes	Yes	Yes
Fit statistics				
Observations	2,664	2,664	2,664	2,664
R^2	0.93148	0.93147	0.93083	0.93126
Within R^2	0.01016	0.01001	0.00086	0.00705

Clustered (state) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Peak crunch period coefficient only. All specifications include state and month-year FEs. Column (3) uses only mobile money and USSD/mobile banking (the two categories most directly substitutable for cash during the crunch). Clustered SEs at the state level.